# Anna Chrzanowska, PhD


Neuroscientist

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## Summary of The Scientific Background


Experienced neuroscientist with 10+ years of expertise in behavioral systems neuroscience. Specialized in a range of rodent behavioral assays, neural circuits genetic manipulations, and neural activity recordings (both *in vivo* freely moving and head-fixed electrophysiology). Skilled in experimental setup design, optimization, and troubleshooting, as well as performing microsurgeries. Demonstrated success in securing fellowships, admissions to competitive training programs and travel grants. Proven leadership in mentoring Master's and Bachelor's students. Actively involved in organizing symposia and conferences, and in science popularization. Engaged member of the international community of young neuroscientists.


## Skills

**Technical Skills:** Rodent behavioral assays, neural circuits genetic manipulations, *in vivo* electrophysiology (freely moving & head-fixed), two-photon calcium imaging, microsurgeries, brain histology, viral injections, optogenetics, chemogenetics, patch clamp recordings (*in vitro*).

**Engineering & Software:** Experimental setup design, optimization, troubleshooting, Bonsai reactive programming, Arduino, electronics, virtual reality environments for mice.

**Professional:** Mentoring, project supervision, scientific writing, manuscript editing, organizing symposia/conferences, science popularization.

## Education

**PhD** **Arenberg Doctoral School, KU Leuven Department of Science**, Biology — Leuven, Belgium, Jan 2018 – Aug 2023

- Dissertation: Unraveling the cell type specific underpinnings that shape collicular mediated visual behaviors
- Supervisor: Dr. Karl Farrow (Neuro-Electronics Research Flanders NERF)

**Master's Degree** **Jagiellonian University**, Neurobiology — Cracow, Poland, Oct 2015 – Nov 2017

- Dissertation: Circadian analysis of the orexin fibers distribution in the rat lateral geniculate nucleus of the thalamus
- Supervisor: Prof. Dr. Marian H. Lewandowski

**Bachelor's Degree** **Jagiellonian University**, Neurobiology — Cracow, Poland, Oct 2012 – July 2015

- Dissertation: Role of GABA-ergic innervation in the mammalian biological clock
- Supervisor: Prof. Dr. Marian H. Lewandowski

## Research Experience

**Bassem Hassan lab, ICM – Paris Brain Institute**, Postdoctoral Research — Paris, France, Jan 2024 – present

Aim: To investigate links between brain wiring structure and physiology and their significance for behavior.

- Neurodevelopmental fly genetics
- Behavioral experiments
- *In vivo* two-photon calcium imaging on behaving flies

**Marion Silies Lab, Johannes-Gutenberg-University of Mainz**, Visiting Scientist — Mainz, Germany, Mar 2025 – Mar 2025

Aim: Data analysis of calcium imaging signals

**Karl Farrow lab, NERF – empowered by imec, KU Leuven and VIB**, PhD Research — Leuven, Belgium
Aim: To investigate the role of distinct neuronal cell types in visually-guided innate behaviors. — Jan 2018 – Aug 2023

- Open-field behavioral paradigms to study innate behaviors in mice
- *In vivo* freely-moving and head-fixed Neuropixels recordings in mice
- Cell-type specific optogenetics and chemogenetics
- Viral injections

**Jagiellonian University (under supervision of Dr. Marian H. Lewandowski)**, Undergraduate Research — Cracow, Poland — Oct 2012 – Nov 2017
Aim: To investigate circadian changes of orexinergic innervation in rats' visual system.

- Brain histology
- Assistance for *in vitro* patch clamp recordings

**Karl Farrow lab, NERF**, Internship — Leuven, Belgium
Aim: To establish a behavioral protocol for visually-triggered orienting and defensive behaviors in a virtual reality environment. — Mar 2017 – July 2017

- Virtual reality environment for mice
- Head-fixed behavioral protocols

**Behavioral Neurodynamics Lab (Ponomarenko/Korotkova Lab), Leibniz-Institute for Molecular Pharmacology (FMP)**, Internship — Berlin, Germany — Oct 2016 – Feb 2017
Aim: To investigate sleep and memory regulation by forebrain circuits.

- Projection- and cell-type specific optogenetic interrogation
- Local field potential recordings *in vivo*
- Freely-moving behavioral protocols

**Laboratory of Neurodegeneration (Kuźnicki Lab), International Institute of Molecular and Cell Biology**, Internship — Warsaw, Poland — Sept 2016
Aim: To investigate the role of calcium-related proteins involved in Huntington's disease.

- Development of *in situ* proximity ligation assay (PLA) on brain slices

**Laboratory of Cell Biology of the Synapse (Matteoli Lab), Neuroscience Institute, CNR**, Erasmus+ Traineeship Program — Milan, Italy — Jan 2015 – Mar 2015
Aim: To investigate the role and molecular mechanisms of Eps8, the actin regulatory protein in synapse plasticity.

- Western blot analysis

## Awards and Distinctions

- Carnot Training Grant for short collaborative visits (2025)
- Marie Sklodowska-Curie Actions Postdoctoral Fellowship (2025)
- The French Foundation for Medical Research (FRM) Postdoctoral Fellowship (2024)
- Fyssen Foundation Postdoctoral Fellowship (2023)
- Participant of Transylvanian Experimental Neuroscience Summer School (TENSS) (2021)
- Participant of The Cajal NeuroKit Course (2021)
- Research Foundation – Flanders (FWO) PhD Fellowship (success rate ~20%) (2018 - 2022)
- Travel Grant to attend The CAJAL Advanced Neuroscience Training Programme (2018)
- Travel Grant to attend 7th Neuron Technology Summer School (2017)
- The Deans' Award for Outstanding Academic Achievement (2015/2016, 2016/2017)
- Laureate of the competition "Grasz o staż" ("Win the internship") (2016)
- Organizing Committee Award for the Best Theoretical Poster – 5th Aspects of Neuroscience (2015)
- Scientific Committee Award for the Best Theoretical Poster – 4th Aspects of Neuroscience (2014)

## Publications

**Rethinking naturalistic behavior: The Terzolas Manifesto** 2025
**Anna Chrzanowska**, Mateusz Kostecki, Natalia Krasilshchikova, Matilde Perrino, Luigi Petrucco
10.31219/osf.io/ch3an_v2 (The Open Science Framework)

**Multiplexed Surface Electrode Arrays Based on Metal Oxide Thin-Film Electronics for High-Resolution Cortical Mapping** 2023
Horacio Londoño-Ramírez, Xiaohua Huang, Jordi Cools, **Anna Chrzanowska**, Clément Brunner, Marco Ballini, Luis Hoffman, Soeren Steudel, Cédric Rolin, Carolina Mora Lopez, Jan Genoe, Sebastian Haesler
10.1002/advs.202308507 (Advanced Science)

**Optogenetic fUSI for brain-wide mapping of neural activity mediating collicular-dependent behaviors** 2021
Arnau Sans-Dublanc, *Anna Chrzanowska***, Katja Reinhard, Dani Lemmon, Bram Nuttin, Théo Lambert, Gabriel Montaldo, Alan Urban#, Karl Farrow#
10.1016/j.neuron.2021.04.008 (Neuron)

**Research culture: science from bench to society** 2021
Lorenzo Canti, **Anna Chrzanowska**, M. Giulia Doglio, Lia Martina, Tim Van Den Bossche
10.1242/bio.058919 (Biology Open)

**Multiple excitatory actions of orexins upon thalamo-cortical neurons in dorsal lateral geniculate nucleus – implications for vision modulation by arousal** 2017
Lukasz Chrobok, Katarzyna Palus, **Anna Chrzanowska**, Mariusz Kępczyński, Marian Lewandowski
10.1038/s41598-017-08202-8 (Scientific Reports)

**Disinhibition of the intergeniculate leaflet network in the WAG/Rij rat model of absence epilepsy** 2017
Lukasz Chrobok, Katrzyna Palus, Jagoda S Jeczmien-Lazur, **Anna Chrzanowska**, Mariusz Kepczyński, Marian Lewandowski
10.1016/j.expneurol.2016.12.014 (Experimental Neurology)

**The role of collicular cell types in behavioral reactions to different threat intensities in mice**
Bram Nuttin, *Anna Chrzanowska***, Amber Bruynsteen, Arnau Sans-Dublanc#, Karl Farrow#
(In preparation)

## Conference Presentations

- **Invited Speaker:** How to write a successful MSCA proposal; PORT for Health Neuroscience, Wroclaw, Poland (2025)
- **Invited Speaker:** Cell vibes: how collicular cell-types contribute to guiding mice innate behaviors; Neurons in Action, Warsaw, Poland (2025)
- **Oral Presentation:** Towards embracing brain complexity: lessons from cell types of the superior colliculus; ENCODS Paris, France (2022)
- **Oral Presentation:** A cell-type specific understanding of innate defensive behaviors: the collicular perspective; Neuromatch 3.0 (2020)
- **Poster Presentation:** Neural circuits of individuality in the context of learning; VIP & DIF 2024, Paris, France (2024)
- **Poster Presentation:** Locomotion shapes visually-evoked neural dynamics through modulation of collicular populations; COSYNE, Lisbon, Portugal (2024)

## Student Supervision

**Master Students:** Helin Polat, Hala Rachel El Hajj (2024-2025); Chaimae Ouriaghli Taouil (2021-2022); Liuyin Yang, Nidaa Elsayed (2019-2020); Carolina Tigre Avelar (2017-2018)

**Bachelor Students:** Sofia Rebiha Decesaro (2024); Diana Dayekh (2023-2024); Martyna Palys, Sade Adeyemi (2022-2023); Diethe Peeters, Tom Voet, Mathis Van den Bruel (2021-2022); Mathilda Corcoles (2020-2021)

## Major Collaborations

- **Alan Urban, NERF:** Established novel technology to study cell-type specific brain-wide circuits in mice (opto-fUSI).
- **Nanoengineering team of imec:** Collaborated to test *in vivo* a novel device for large-scale brain recordings.

## Contribution to Scientific Community

- Teaching assistant at Transatlantic Behavioral Neuroscience Summer School (2021 – present)
- Local teaching assistant at NERF for the Cajal NeuroKit Course (2022)